Ibasho 居場所の向上

Status, explainer and walkthrough

Progress

Milestone	State	Notes
Research and plan	DONE	Eight research briefs, the plan of record, three independent reviews. M1 and M2 reports in `report/`.
G0 · Scaffold	DONE	Monorepo, shared contracts, layer and ownership rules, CI.
M-0 · Walking skeleton	DONE	The reference home runs through every layer: physics, rules, search and money, Lean checker, narrator, scene, agent link and the first app screen.
M-A · First light	ACTIVE	Full-strength physics and validation data, more of the design system, onboarding and the reveal with the 3D house.
M-B · Three futures	ACTIVE	Full UK rules, score and EPC band, payback, the “If nothing is done” line and negatives on every result.
M-C · Sealed	ACTIVE	Checker now also covers bills, score and completed measures. Proofs page still to build.
M-D · Story	TODO	Five chapters, calibration from bills, custom what-if requests, welcome back.
M-E · Release	TODO	Landing posters, accessibility and performance passes. Testing stays light until then.
v3 plan	DONE	A studio for eight building types in a real 3D city, Hyderabad first. Plan in `docs/PLAN-V3.md`.
V-0 · Contracts v3	TODO	Shared types for the building model, results, stages, settings and goals; new packages; installs.
V-P · Prototype	TODO	Every flow (new, revisit, restart, what-if, update) on phone, tablet and desktop, on preview data with an honest staged progress bar.
V-1 to V-6 · Real stages	TODO	Geometry, sun and radiation, daylight, energy, codes and ratings, optimisation, each replacing its preview in turn.
V-7 · Partner and memory	TODO	Memory across buildings, typed goals, live recommendations, a design partner that proposes and never computes.
V-H · Hyderabad hero	TODO	Vindhya Bhavan at IIIT Hyderabad, in its city, end to end and offline.

What Ibasho is

Ibasho is a digital twin of a home that you can edit. Change the house and the model recomputes it, using a documented reduced-order heat balance [1], [2]. It searches the retrofit packages you can actually build and shows the best few. Lean 4 [3] proves the bookkeeping. An AI narrator explains results but cannot change a number. The visual language takes cues from architectural storytelling [4].

Direction v3

From October 2026 Ibasho grows from one home into a studio for many kinds of building: offices, hotels, schools, hospitals, labs, apartments, rentals and single family homes. You place a building in its real city, draw it floor by floor, and see its sun, shadows, daylight, energy by end use, carbon, bills, code standing and green rating path, with the cheapest bundle of changes that meets a goal. Hyderabad comes first, with a building on the IIIT campus as the hero; the London terrace stays. The same principles hold: numbers come

from documented models, unknown is not zero, and Lean checks the bookkeeping. Work runs outside in: a full prototype of every flow on clearly labelled preview data first, then each stage made real in turn.

Features

Feature	State
G0 scaffold: a 28-package monorepo with enforced layers and ownership. The kernel holds every contract plus exact fixed-point money, exact fractions for the score, SHA-256 hashing [5], a seeded generator [6] and Latin hypercube sampling [7], all tested. Both apps boot in Chromium and Firefox.	DONE
Andon UI kit, first slice: day and night colour tokens checked for contrast and colour blindness [8], [9], three self-hosted font families, seal kanji baked from Yuji Syuku [10], and the number chip, check seal and proved claim seal.	DONE
Andon UI kit, futures and studio parts: a sliding paper sheet with snap points, the narrator strip, the score with four groups (a band while facts are missing, never a letter), the estimated EPC chip with its spread, the three futures folding screen with what gets worse on every card and an “If nothing is done” line, measure cards, per-fuel diverging bars, a lattice slider, a twelve-month ring, the why-not list and the “possible after a check” lane with source grades. Every chart has a table, and every number opens its evidence.	DONE
Physics, walking-skeleton slice: GH1, a 1905 London terrace, runs through a steady monthly heat balance [2], [11]. It models fabric and ventilation losses, solar and internal gains, hot water, boiler efficiency and a fixed heat pump COP, all on London climate data [12]. Three measures (loft, draught-proofing, heat pump) patch the twin, and a package is always recomputed whole. Maths is deterministic, ported from fdlibm [13]. Energy is rounded to whole watt hours exactly once. The fast path for heating changes matches a full rerun bit for bit.	ACTIVE
Hero assets: four Poly Haven skies [14], four material sets and 38 pieces of Kenney [15] and Quaternius [16] furniture for the GH1 terrace, all CC0. Skies ship as compact gain maps [17], models in metres with meshopt. Every file is listed with its licence and checksum, and the credits page is generated from that list. 5.8 MB of the 8 MB budget.	DONE
Rules, walking-skeleton slice: rule conditions are data, evaluated in three-valued logic [18], so an unknown fact is never read as “no”. A package blocked only by something we do not know goes to “possible after a check”, never to the front. Two rules run on GH1: the budget, which does nothing until a budget is set, and a heat pump on a possibly shared electricity supply, which waits for the network operator to approve. A randomised test checks that an unknown block is never feasible.	ACTIVE
Rules for the three futures: the full English rule set for the hero home, read as plain sentences. Permitted development for heat pumps, solar panels and wall insulation [19], [20] gives a planning application, never a hard no, and the heat pump noise limit of 37.0 dB(A) [21] is decided with exact fractions that bracket every logarithm, so the answer is yes or no only when every rounding agrees. Building Regulations [22], [23] add what a measure obliges: walls to 0.30 when more than half is renovated (0.70 is the trigger, not the target), emitters for 55 C, ventilation by the AD F table, and a costed moisture check before inside wall insulation on solid brick. Grid export follows G98 and G99 [24], landlords see the rental rules [25], and measures that cannot apply (a cylinder jacket on a combi) are ruled out. Every rule has a source, a grade and its own test; old rules retire on their date.	DONE
Search and finance, walking-skeleton slice: GH1′s 8 packages (loft, draught-proofing, heat pump) are enumerated in full and costed. Bills are exact at the October 2026 price cap [26], with standing charges. Carbon is counted per fuel and month with SAP 10.2 factors [2]. Grants include the Boiler Upgrade Scheme [27]. Upfront cost and NPV at the Green Book 3.5% [28] are in whole micro-pence with named rounding. The best options form a front, every other option has a named witness that beats it, and the three picks follow the stated tie-breaks. The search	ACTIVE

runs in a worker, reports progress and stops on cancel. It emits the certificate input for the Lean checker, and the same request always gives the same hash. A property-value table [29] is ready but not yet shown.

Search and finance, first M-B pass: every package shown now lists what gets worse beside what gets better (a fuel or bill that rises, a negative return, more carbon, a lower score or band, a moisture or noise risk), each backed by evidence. Paybacks, a rate of return reported only when it is unique, and a 0% loan with a "cash positive from month 1" check [28]. The Green Home Score is computed in exact fractions from the Lean constants, with four groups; it shows a range until comfort hours land. The estimated EPC band uses the rating run at RdSAP 10 prices [2], [11] and always shows one band either side. Flip points find every price or discount rate where a pick changes. "If nothing is done" ages the home with documented decay models, year by year, with costs and carbon. ACTIVE

Lean checker and verify, walking-skeleton slice: a checker in Lean 4 [3] re-derives every saving, carbon figure, upfront cost, NPV, rule status, front, witness and pick of a certificate from the physics and pack data, and a machine-checked theorem says that passing every check means the claims hold. The constants the app uses, the certificate types and the claim seals with each theorem's statement are generated from Lean, with a SHA-256 spec hash [5], and CI fails if they drift. A TypeScript mirror gives the same verdicts. On GH1 both checkers pass the real certificate and both catch three planted errors: a flipped saving, a rejected measure on the front, and a heat pump with an unknown supply marked feasible. DONE

Lean checker, gaps closed: the checker now also re-derives every annual bill from the tariff, recomputes the Green Home Score of every package on show in exact fractions, and treats a measure you marked done like one you ruled out, so it never comes back. The proof that passing every check means the claims hold covers all three. On GH1 both checkers pass the real and a scored certificate and catch a done measure on the front, a wrong score total and an unscored pick. DONE

Words and landing page: plain guidance for all 16 home measures (how many days of work, which rooms, whether you can stay in, who to call, which accreditation to look for, and three questions to ask before you sign) [30], [31], [32], [33], a glossary of about 50 terms, and a voice checker that rejects "guaranteed" and "save up to". The landing page paints as static HTML in five chapters, and a chapter appears only once the feature it describes exists. DONE

Lean spec, first slice: the bookkeeping rules are written once in Lean 4 [3] and proved for every input. Exact rounding to the nearest even unit, discounting, per fuel and per month savings and carbon, the score (weights add to one, always 0 to 100, never lower for a better home), three-valued rule logic [18] (an unknown is never read as a pass), the heat pump noise test of MCS 020 a) [21] decided only when rounding cannot change it, the Pareto front (nothing better is hidden), the three picks, and memory (a measure you ruled out never comes back). The build fails if any proof is missing or uses an extra axiom. ACTIVE

Agent link, first slice: an AI agent such as Claude Code can explore a home through 19 MCP tools [34], headless or live in the open app. It reads numbers only as evidence with their ids, changes a home only with typed operations, and records memory only in the person's own words. A raw number in its narration is refused, following proof-carrying numbers [35]. The live link is local-only, with a per-launch token and a one-time pairing code. It installs into Claude Code as a plugin [36], or as a skill, subagent and hook, and uninstalls without a trace. Headless numbers now come from the real engine: a search through the agent link gives the same result, hash for hash, as the app's engine on the reference home. ACTIVE

Grounded narration and memory: the story of a home is told from a template that is the same byte for byte on every run, with numbers only as evidence chips. A local model on your computer (Ollama [37]) may polish the words, but every sentence is checked by eight grounding rules and falls back to the template if it writes a number, misreads a sign, recommends something you ruled out or copies a note. Notes from previous owners are read ACTIVE

by a quarantine that keeps only typed facts; text sent to a model is datamarked [38] and logged in an outbound ledger. 60 injection notes plant 0 numbers and 0 memories. The memory of what you told us replays 500 random sessions without ever offering a ruled-out measure, and forgetting deletes for real after 30 days. An optional companion [39] on your machine adds the Lean checker and an Ollama bridge [40].

The app, walking-skeleton slice: the sample home opens at `/app/home/gh1`. The engine works in the background while the house is drawn in pencil; then it is inked, and three futures appear: least upfront, best return and lowest running carbon. Every figure can be tapped to see how we know it. A second checker redoes the sums and the page shows a Checked seal only when it agrees, next to the proved promise that anything you rule out never comes back. Results are kept on your device only. ACTIVE

The app, first light: a new visitor is asked five quick questions (where, which home type, a few pictures, who lives there, then the reveal) in about two minutes, with no money question until the futures appear. Nothing is looked up online. Anything not answered stays an estimate and is marked as one. The reveal opens the home on the 3D stage, drawn on paper by day or lantern-lit at night, with the roof lifting once the sums are done; without 3D the house is drawn in pencil and then inked. Unanswered values come from RdSAP defaults [11]. ACTIVE

Credits: one acknowledgements file names every library, ported algorithm, data source, asset and idea Ibasho builds on, with its licence. Every agent adds to it. DONE

Bad outcomes, contracts: a home can now be shown getting worse. Packs describe how a home ages when maintenance is ignored (a boiler losing efficiency, draught-proofing wearing out, a boiler replacement falling due), each with its sources, its grade and a low to high range. The engine contract has room for an “If nothing is done” line beside the three futures, and every package lists what gets worse beside what gets better. Fuel price scenarios and a confirmed boiler replacement are now part of the owner’s inputs. ACTIVE

The living drawing, first light: your home is built from its survey as a cutaway miniature on paper, roof lifted, in the manner of the fukinuki yatai picture scrolls [41]. It renders with three.js [42] through React Three Fiber [43], by day (Washi) or by lamplight (Yoru), with sun from SunCalc [44]. It redraws only when something changes. Heat leaving the house is shown as drifting motes, a lens shows modelled surface warmth and the living area is tinted by temperature. A plain drawing takes over without WebGL. Every part of the house can be picked, and a text outline lists the same parts for screen readers. Each retrofit measure has its crest and fold, and each kind of home its thumbnail. ACTIVE

City stage, prototype: a building placed in a paper model of its city, lit by the real sun [42], [44]. Zoom from the city to a site, the building, a floor and a room; scrub the day and the year; sweep dawn to dusk over the IIIT campus. Neighbours whose heights are estimates cast a solid shadow at the likely height and a hatched band out to the plausible one, and a typed height closes it [45]. Lenses paint sun hours, shadow hours, radiation that rises up the walls and daylight on a floor that slides out like a drawer, with legends that say Preview. A sun bent variant grows a solar canopy and shading fins [46], and at night only modelled buildings glow. Find a building, floor or room and the camera flies to it. Picking is fast [47], and without WebGL a plan drawing takes over. It now runs on the OpenStreetMap city pack and the traced Vindhya plan, with Andon controls laid out for phone, tablet and desktop. At night the building glows as the lantern of the scene; a radiation rose opens over the roof; a sun bent strip steps through searched designs; adding floors shows whose winter sun you take. ACTIVE

v3 contracts: a building is now a document of storeys, rooms, walls, windows and shades in whole millimetres, placed in a city. Seven analysis stages (geometry, sun, radiation, daylight, energy, codes, optimisation) share one contract whether their numbers are Preview or real, and every Preview number carries its stamp. Settings, goals, code and rating rows, design moves and place search have typed contracts with schemas and round-trip tests. DONE

Plan kernel and golden buildings: exact areas in square millimetres, shared walls found and paired, facades sorted by true orientation, window ratios and shading projection factors, auto zoning and polygon operations [48]. Vindhya Bhavan is rebuilt from the campus floor plans [49] as 93 named rooms (the library, data centre, bank, seminar halls, auditorium, labs and studios) over a cellar and three floors, fitted to its mapped outline [45], with Himalaya, a HITEC City office, a Gachibowli apartment block and the London terrace beside it. Unknowns such as glazing and wall build-ups are listed, never filled. A campus, building, block, floor or room can be found by name, nickname or a near miss (“IIIT”, “Vindhya floor 2”, “KCIS”, “Auditorium”). ACTIVE

Preview results: all seven stages run on any building, edited or not, and stream rough, refined and final results with time left and a cancel that answers within one slice. Numbers come from a plain model on the Hyderabad weather year [50]: end uses add up to the energy use intensity exactly, bills follow the Telangana tariffs to the paisa (per kVAh with time of day for the campus, and the 200 unit cliff for flats), code pre-checks never claim compliance [51], and 256 design options form a front where every option left out has a named better one. Everything is stamped Preview until the real stage lands. ACTIVE

City data: the IIIT campus and two kilometres of Gachibowli as 1,036 mapped buildings and 68 trees in 72 KB [45], [52], rebuilt byte for byte from a cached extract. Every building has a height range and says where it came from; most heights are unknown and are drawn as ranges. A small London pack surrounds the terrace. ACTIVE

Real energy, first slice: every room is simulated hour by hour through a typical year with a five-node heat model [53], driven by the Hyderabad or London weather year [50], the room’s use (people, lights, equipment and fresh air [54]), and the plant that serves it: split or VRF units, chilled water with its pumps and fans, ceiling fans first in homes [55], boilers or heat pumps in the UK. It gives cooling, heating, lighting, equipment, fans, pumps, hot water and solar per month, the energy use intensity, bills and carbon, in well under a second, and each sample building lands inside its published benchmark band [56]. ACTIVE

Offices of interest and design before you build: the Lloyds Technology Centre, Qualcomm and Vanguard offices in Hyderabad appear when you search for them, each in the tower that houses it, drawn from mapped outlines and public facts [45], [57], [58], [59], with floor plates marked as estimates. A new building can be designed and placed on real plots at King’s Cross in London or in Hyderabad, and checked against the rules of that place: Part L, the Future Homes Standard, overheating and the London Plan in England [22], [60], [61], or ECSBC in India. Targets such as net zero ready [62], LEED Gold or ECSBC Super each show met, not met or unknown with a source. Moving a building on the map reuses its drawing and reruns only the sums that change. ACTIVE

Studio kit: a seven-bead progress bar that shows rough, refined and final results with time left and a cancel; a Preview stamp that is never a seal; a scale ladder from city to room; mode tabs; one panel for every analysis lens with a legend that states unit, resolution, period, method and “Modelled”; colour ramps interpolated in OKLab [8]; eight building type cards with crests and daily schedules; an energy use intensity gauge, an end-use bar and a ladder that puts GHMC stars, ECSBC tiers, LEED, IGBC and EDGE on one savings axis; settings rows that say where a value came from and how many numbers it changes; a compare grid and a version film strip; and Andon inputs (a lattice range, a 24 hour sun scrubber, a list picker) so no browser control shows. DONE

Plan editor: draw rooms as rectangles or polygons, split and merge them, paint room functions, set windows by share of wall and shading by preset (chajja, deep overhang, fins, jaali), stack and repeat storeys, trace over an image of a plan, with snapping, typed sizes in metres or feet and inches, live window and projection factor meters, overlap and untyped rooms drawn on the plan, keyboard and touch, and undo and redo as Immer patches [63]. ACTIVE

Studio, prototype of every flow: start a place, pick a type, start from a template or the Vindhya sample, place it among the mapped neighbours and see first results; come back to what changed since the last visit, with old and new figures side by side; start over or branch a place without losing the old one; compare the building as it is with up to three what-ifs where every lever recomputes the whole variant; and edit the building or a rate and watch only the touched stages rerun, with stale numbers hatched. Find any building, floor or room by name. Everything runs on desktop, tablet and phone, and every number is stamped Preview. ACTIVE

Design before you build, placing: drag a designed building onto a plot in the city, turn it and set its ground level; the stage flags overlaps with existing buildings and names the neighbours that shade it, and shows two or three plots or orientations side by side with the same view. At night the modelled building is the brightest lantern, and estimated heights are hatched only where they can shade it [45]. ACTIVE

Design before you build: start from a brief with typed goals (net zero ready, LEED Gold, ECSBC Super, Part L and the Future Homes Standard, lowest running cost), draw it from a template or as massing with cores, windows by façade and overhangs, place it on the Hyderabad or London map by clicking open ground, turning it and checking overlaps with mapped buildings, compare up to three plots and orientations side by side, see where it stands against each goal (met, not met or unknown, with its source), apply the cheapest changes and save it as its own place. Every number is Preview until calculated for the building. ACTIVE

v3 landing page and studio words: the landing now introduces a studio for any building in its real city, for homeowners, architects, campus estates teams, developers and businesses. Each feature section (city, sun and shade, daylight, energy and bills, codes and ratings, comparing futures, designing before you build) appears only once the studio can show it, marked Preview until that step is calculated. A plain glossary covers the new terms (EUI, sDA, ASE, ECSBC, ENS, Part L, LEED points), with studio wording for progress, code pre-checks and the eight building types, and a checker that keeps development notes out of customer copy. DONE

Real site analysis, first stages: sun position from the NREL Solar Position Algorithm [64] on deterministic maths, hourly weather for Hyderabad and London, an all-weather sky [65] summed per month [66], and a ray-traced visibility kernel over the building and its city context. It gives sun hours, sky view, monthly radiation with a shading ledger that closes exactly (who takes your sun, including the building itself), and a fast daylight estimate per room (ASE exact from direct sun, sDA simplified) [67]. Same results as the Preview stages, so swapping is one line; rough results in under a second on the campus buildings. ACTIVE

Codes and ratings, real: the design is checked against the codes that apply where it stands. In India, the commercial energy code by its prescriptive limits and by the ratio of its energy use to a reference design run on the same engine, the residential envelope rules, window openings and the coming rating rule; in England, Part L 2021 against the notional building, the Future Homes Standard, the overheating check and the London Plan; everywhere, the ASHRAE 90.1 Appendix G baseline [68]. One savings axis carries the ladders: GHMC stars [69], LEED points [70], IGBC, GRIHA, EDGE and BEE office stars. Each answer is met, not met or unknown, with its clause and source; values not given stay unknown, never a guess. ACTIVE

Walkthrough

Written once the app runs end to end.

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